DSA06 - Data Handling and Visualization | Activity 5: Normality Detection using Q-Q Plots | Slot C

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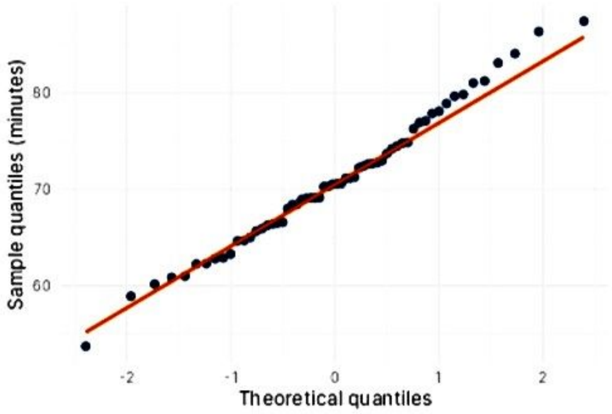
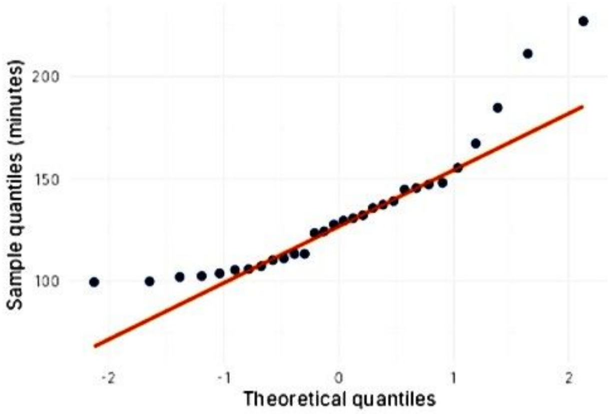
Each chart is a normal Q-Q plot made in R (stat_qq and stat_qq_line; red line = perfect normal fit). Part A (16 marks): for every chart decide whether the data are normal and name the shape from this list: Normal, Right-skewed, Left-skewed, Heavy-tailed, Light-tailed, Bimodal, Normal with outlier(s); then cite the evidence you see.

Chart 1: lab experiment duration (minutes), n = 60



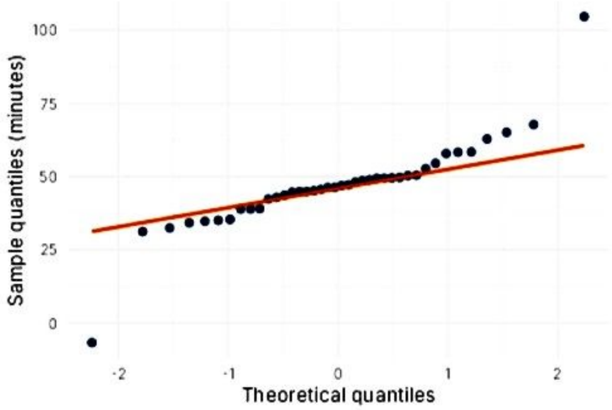
Normal? (Yes / No)	
Shape (from the list)	
Evidence in the plot	

Chart 2: daily study time (minutes), n = 30



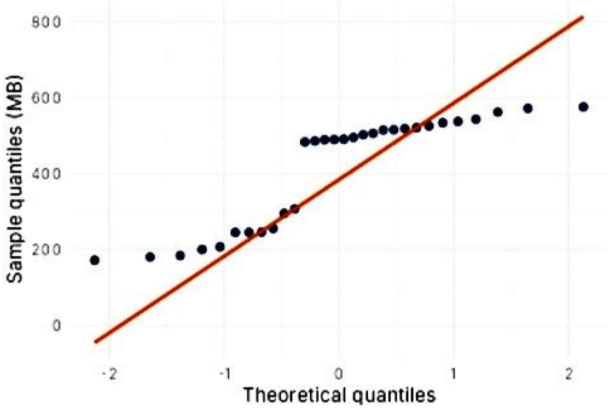
Normal? (Yes / No)	
Shape (from the list)	
Evidence in the plot	

Chart 3: commute time (minutes), n = 40



Normal? (Yes / No)	
Shape (from the list)	
Evidence in the plot	

Chart 4: mobile data used per day (MB), n = 30



Normal? (Yes / No)	
Shape (from the list)	
Evidence in the plot	

Part B (4 marks)

B1. Chart 3 departs from normality. Name the departure and state the remedy you would try before using mean-and-SD methods (for example log transform, square, split the data, robust methods). [2]

B2. Fill the blanks to draw a Q-Q plot with its reference line for a data frame df with a column value: ggplot(df, aes(sample = value)) + ________ + ________ [2]

Activity-5
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Part A: Observation Table:-

1) Chart 1: lab experiment duration (Minutes), n=60

Normal? Yes

Shape: Normal

Evidence: most Points are close to the reference line.

2) Chart 2:- Daily study time (Minutes), n=30

Normal? No

Shape: Right skewed

Evidence: The Points curve upward and move away from line at the right end

3) Chart 3:- Compute time (Minutes), n=40

Normal? No

Shape: Right-skewed

Evidence:- One very low value and one very high value are far from the line.

4) Chart 4:- Mobile data used per day (MB), n=30

Normal? No

Shape: Left skewed / Non-normal

Evidence: The most Points dono follow the reference line and form a curved Pattern.

Part B: Short Answers:-

B1. Chart 3:- departs from normality. Name the Depature and remedy.

* Departure :- Right skewed data with outliers

* Remedy: Apply a log transformation and Check the before using Mean and SD.

B2 Fill in the blanks to draw a Q-Q-Plot:-

```
ggPlot(df, aes(sample= value))+ stat-qq()+
stat_qq-line().
```